AGRICULTURAL DRIP IRRIGATION FIELD PLUMBING & EXECUTION MANUAL

2-Acre Commercial Moringa Farm (PKM-1 & Bhagya Cultivars)

DOCUMENT CONTROL & METADATA

- **Project:** 2-Acre High-Density & Commercial Moringa Irrigation System
- **Location:** Agricultural Farm Sector
- **Engineering Standard:** IS 12786 (Irrigation Laterals) / IS 4985 (PVC Pressure Pipes)
- **Authoring Suite:** ArtCraft Engineering Toolchain (CADCraft, VectorCraft, GridCraft, WordCraft)
- **Document Version:** Revision 1.0 (Field Approved)
- **Scope:** Head Unit, Mainline Network, Overhead Double-Tee Risers, Inline & Online Laterals, Flushing & Commissioning

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1. PROJECT OVERVIEW & HYDRAULIC DESIGN SUMMARY

This manual governs the complete field installation of the pressurized drip irrigation infrastructure serving 2.0 gross acres divided into two distinct agro-horticultural zones:

Zone Breakdown

1. **Zone 1: Solar Borewell & Mainline Distribution Network**
    - **Water Source:** 5.0 HP DC Solar Submersible Borewell at coordinate `(X=590, Y=20)`
    - **Operating Pressure:** 2.5 kg/cm² (35.5 PSI) dynamic delivery pressure
    - **Discharge Rate:** 18,000–22,000 Liters Per Hour (LPH) depending on solar irradiance
    - **Mainline:** 2" (63mm OD) PVC Class 2 (6 kg/cm² pressure rated) solvent-weld network
    - **Total Mainline Run:** 730 ft installed (80 ft North + 90 ft West to Tee A + 215 ft West to Tee B + 145 ft North Plot A + 240 ft North Plot B)
2. **Zone 2: Plot A (East Field) — PKM-1 Annual Drumstick**
    - **Net Planting Area:** 1.0 Acre (390 ft East-West x 130 ft North-South)
    - **Crop Spacing:** High-density 3 ft plant-to-plant along trench x 5 ft between trenches

- **Total Population:** 3,000 Moringa trees (leaf meal & pod production)
- **Riser Configuration:** 4 Overhead Double-Tee Stations (A1–A4 spaced at 30 ft intervals)
- **Subline:** 1.5" (50mm OD) Class 2 PVC (120 ft total run, 8 arms of 15 ft each)
- **Lateral Specification:** `16x4x40 Inline Pipe` (16mm LLDPE with integrated 4 LPH labyrinth drippers spaced every 40cm); 24 trenches x 390 ft = 2,926m installed
- **Take-off Connection:** `50x16x2 Way Take-Off Collar` (V-Connector) feeding East & West simultaneously

3. **Zone 3: Plot B (West Field) — Bhagya (KDM-01) Drumstick**
   - **Net Planting Area:** 1.0 Acre (165 ft East-West x 260 ft North-South)
   - **Crop Spacing:** Commercial orchard 7 ft plant-to-plant x 8 ft between rows
   - **Total Population:** 750 Moringa trees (large fruit/export quality)
   - **Riser Configuration:** 4 Overhead Double-Tee Stations (B1–B4 spaced at 64 ft intervals)
   - **Subline:** 1.25" (40mm OD) Class 2 PVC (256 ft total run, 8 arms of 32 ft each)
   - **Lateral Specification:** Plain 16mm LLDPE pipe; 32 rows x 165 ft = 1,610m installed (Westward run only)
   - **Emitter Specification:** `Tunning Dripper (P)` (0–70 LPH adjustable turbulent flow) punched on-site every 7 ft (768 drippers total)
   - **Take-off Connection:** `40x16x1Way TF Collar` pointing West

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2. EXCAVATION & TRENCHING SPECIFICATIONS

Precise earthwork ensures pipeline longevity, prevents mechanical damage from inter-cultivation tractors/rotavators, and guarantees freeze/heat insulation.

Pipeline Element	Nominal Size	Minimum Trench Depth	Minimum Trench Width	Bedding Material
Zone 1 Mainline	2" (63mm) PVC	2'-6" (750 mm)	1'-6" (450 mm)	2" sifted sand or loose soil
Plot A Sublines	1.5" (50mm) PVC	2'-0" (600 mm)	1'-2" (350 mm)	Fine rock-free soil bed
Plot B Sublines	1.25" (40mm) PVC	2'-0" (600 mm)	1'-2" (350 mm)	Fine rock-free soil bed
Plot A Trenches	24 Trenches	0'-8" to 1'-0" (250 mm)	1'-0" (300 mm)	Organic compost & soil mix

Pipeline Element	Nominal Size	Minimum Trench Depth	Minimum Trench Width	Bedding Material
Plot B Rows	32 Surface Rows	Surface / 2" furrow	N/A	Surface laid along row stake

Trenching Rules:

1. **Bottom Grading:** Trenches must be free of sharp rocks, shale, roots, and clods larger than 10mm.
2. **Expansion Looping:** Never lay PVC pipes dead straight under tension. Snake the pipe slightly inside the trench (1 ft lateral deflection per 100 ft) to accommodate thermal contraction during winter nights.
3. **Backfill Protocol:** The first 6 inches of backfill must be hand-tamped sifted topsoil. Do NOT run heavy tractors over trenches until 24 hours after full backfilling.

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3. HEAD UNIT & FILTRATION ASSEMBLY (ZONE 1)

Located at coordinate `(590, 20)` directly adjacent to the 5 HP Solar Array:

```
[Borewell Discharge Pipe]
        │
        ▼
[2" Non-Return Valve (NRV)] ── Prevents water hammer back to pump impellers
        │
        ▼
[2" PVC Gate Valve #1] ──────── Primary flow isolation and throttling
        │
        ▼
[2" Filter 'T' Type] ──────────── 120-mesh disc/screen filter (Item #11)
        │
        ▼
[Pressure Gauge Port] ──────── 0–6 kg/cm² glycerin-filled gauge
        │
        ▼
[2" Borewell Tee (590, 20)] ── South outlet capped with 2" End Cap; North feeds Mainline
```

Head Unit Installation Steps:

1. Mount the 2" T-Type Filter vertically or horizontally with the drain/flushing port facing downward.
2. Ensure flow arrow on the filter body points North toward the mainline.
3. Apply 5–6 wraps of Teflon tape on all male threaded transitions before screwing into the filter housing. Do NOT use solvent cement on threaded filter collars.
4. Provide a brick or concrete pedestal under the filter and gate valve to support hydraulic dynamic load.

4. 2" MAINLINE NETWORK LAYOUT & SOLVENT WELDING

Master Alignment Coordinates:

- **Station 1 (Borewell Tee):** `(590, 20)` — South capped with 2" End Cap.
- **Run 1:** Northward run of **80'-0"** from `(590, 20)` to `(590, 100)`.
- **Station 2 (Corner Tee):** `(590, 100)` — East outlet capped with 2" End Cap.
- **Run 2:** Westward run of **90'-0"** from `(590, 100)` to `(500, 100)` (**Tee A**).
- **Run 3 (Inter-Plot Mainline):** Westward run of **215'-0"** from **Tee A** `(500, 100)` to **Tee B** `(285, 100)`.
    - Runs under the West half of Plot A (190 ft) + across the 25 ft buffer roadway.
- **Station 3 (Tee B):** `(285, 100)` — West outlet capped with 2" End Cap.

PVC Solvent Welding Best Practices:

1. **Cutting:** Cut pipes squarely using a fine-tooth hacksaw blade or ratcheting PVC cutter. Remove all internal and external burrs with a deburring chamfer tool.
2. **Dry Fit:** Check dry fit of pipe into fitting socket; it should enter 1/3 to 2/3 of the depth snugly.
3. **Cleaning:** Wipe pipe ends and fitting interior with a dry cotton cloth.
4. **Solvent Application:** Using the provided 3L heavy-duty PVC Solvent Cement, apply a uniform, generous layer around the pipe exterior and a thin, even layer inside the socket.
5. **Assembly:** Push the pipe firmly home with a 1/4 turn twist to distribute the cement evenly. Hold rigidly for 30 seconds to prevent push-out.
6. **Cure Time:** Allow **minimum 2 hours** initial set before moving pipes, and **24 hours full cure** before pressure testing.

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5. 4-FOOT OVERHEAD DOUBLE-TEE RISER MANIFOLD ASSEMBLY

The proprietary overhead double-tee manifold provides waist-height ergonomics (+4'-0" above finished grade), eliminating the need for operators to crawl into muddy pits to turn valves or monitor subline pressures.

Assembly Sequence (Standard for all 8 Riser Stations):

1. **Buried Mainline Tee:** In the mainline trench bottom (-2'-6"), solvent-weld a `2" PVC Equal Tee` with the side outlet pointing vertically (+Z).

2. **Vertical Riser Pipe:** Glue a **4'-0" cut length of 2" PVC pipe** into the tee outlet, extending straight up through finished grade level to +4.0 ft.
3. **Overhead Double-Tee:** Glue a `2" PVC Equal Tee` at the top of the 4ft riser, oriented horizontally with branches facing North and South.
4. **Reducers:**
    - **Plot A Risers (A1–A4):** Glue a `2" x 1.5" (50mm) PVC Bushing Reducer` into both North and South tee outlets.
    - **Plot B Risers (B1–B4):** Glue a `2" x 1.25" (40mm) PVC Bushing Reducer` into both North and South tee outlets.
5. **Ball Valves:**
    - **Plot A:** Solvent-weld a `50mm (1.5") Plain Ball Valve` (Item #13) onto each reducer stub.
    - **Plot B:** Solvent-weld a `40mm (1.25") Plain Ball Valve` (Item #12) onto each reducer stub.
    - *Ensure valve handle operates smoothly and points upwards.*
6. **Overhead Elbow & Drop Pipe:**
    - Glue a 90° PVC Elbow onto the valve discharge pointing directly downward.
    - Solvent-weld a **6'-0" cut length** of drop pipe (1.5" for Plot A, 1.25" for Plot B) down into the subline trench (-2'-0" elevation).
7. **Trench Elbow & Subline Connection:**
    - Glue a bottom 90° PVC Elbow at the trench bottom, turning horizontally into the subline trench.
    - Connect the horizontal buried subline pipe.
8. **Subline Flush Valve:**
    - At the terminal end of every subline arm, install a `50mm Flush Valve` (Plot A) or `40mm Flush Valve` (Plot B).
    - Enclose the flush valve in an 8" PVC sleeve pipe with a removable cap flush with the ground for easy flushing access.

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6. ZONE 2 (PLOT A) EXECUTION: PKM-1 MORINGA INLINE SYSTEM

Spacing & Layout:

- **Mainline Axis:** Runs North from Tee A `(500, 100)` to `(500, 245)` (145 ft total, capped with 2" End Cap at `(500, 245)`).
- **Riser Spacing:** 4 Riser Stations spaced every **30'-0"**:
    - **Riser A1:** `(500, 130)` — Subline arms cover `Y = 115'` to `145'`

    - **Riser A2:** `(500, 160)` — Subline arms cover `Y = 145'` to `175'`
    - **Riser A3:** `(500, 190)` — Subline arms cover `Y = 175'` to `205'`
    - **Riser A4:** `(500, 220)` — Subline arms cover `Y = 205'` to `235'`
- **Subline Arms:** 8 arms total, each running **15'-0"** (30 ft total per station; 120 ft total 1.5" subline).
- **Trenches:** 24 trenches spaced exactly **5'-0" apart** (3 trenches per 15-ft arm x 8 arms = 24 trenches).

Collar Drilling & Lateral Unrolling:

1. **Drilling:** Using a 16mm spade bit or hole saw, drill into the top center of the buried 1.5" PVC subline at exact 5-foot intervals. Clean out all plastic shavings from inside the pipe.
2. **Collar Installation:** Seat the rubber grommet firmly into the hole. Clamp the `50x16x2 Way Take of Coller` (Item #9) around the 50mm pipe, securing the barbed stem through the grommet.
3. **Lateral Laying:** Unroll the `16x4x40 Inline Pipe` (Item #8) carefully along each trench bed. Do not drag the pipe over rough gravel to avoid scratching the LLDPE wall.
4. **Length:** Cut each trench lateral to **390 ft total length** (~195 ft West and ~195 ft East of the central subline).
5. **Connection:** Push the East and West pipe ends firmly over the two barbed outlets of the 2-way collar. Secure with safety lock rings.
6. **End Capping:** At the far East (`X = 695`) and far West (`X = 305`) ends, install `16mm Endcaps` (Item #4) with figure-8 folds.

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7. ZONE 3 (PLOT B) EXECUTION: BHAGYA MORINGA ONLINE SYSTEM

Spacing & Layout:

- **Mainline Axis:** Runs North along the East boundary from Tee B `(285, 100)` to `(285, 340)` (240 ft total, capped with 2" End Cap at `(285, 340)`).
- **Riser Spacing:** 4 Riser Stations spaced every **64'-0"**:
    - **Riser B1:** `(285, 124)` — Subline arms cover `Y = 92'` to `156'`
    - **Riser B2:** `(285, 188)` — Subline arms cover `Y = 156'` to `220'`
    - **Riser B3:** `(285, 252)` — Subline arms cover `Y = 220'` to `284'`
    - **Riser B4:** `(285, 316)` — Subline arms cover `Y = 284'` to `348'`
- **Subline Arms:** 8 arms total, each running **32'-0"** (64 ft total per station; 256 ft total 1.25" subline).

- **Rows:** 32 rows spaced exactly **8'-0" apart** (4 rows per 32-ft arm x 8 arms = 32 rows).

Collar & Dripper Installation:

1. **Collar Installation:** Drill 16mm holes in the 1.25" PVC subline at 8-foot centers pointing **West only**. Install the `40x16x1Way TF Coller` (Item #6) with rubber seal.
2. **Plain Pipe Laying:** Unroll plain `16mm Pipe` (Item #1) along each row for **165'-0" West** (from `X = 285` to `X = 120`). Connect to collar barb.
3. **Dripper Punching:**
    - Measure 7'-0" from the plant base along the lateral.
    - Using the supplied `Punching Tool` (Item #3), punch a clean 3mm hole in the top quadrant of the lateral pipe.
    - Immediately press the barbed base of a `Tunning Dripper (P)` (Item #2) into the punched hole until it clicks audibly into place.
    - Repeat for 24 plants per row (24 x 32 = 768 drippers total).
4. **End Capping:** Fold and secure the West end of all 32 rows with `16mm Endcap` (Item #4) at `X = 120`.

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8. SYSTEM FLUSHING & PRESSURE TESTING PROTOCOL

Flushing removes sand, PVC burrs, cement debris, and insects before water enters sensitive dripper labyrinths. **Flushing must always proceed strictly from largest diameter to smallest diameter.**

```
[STAGE A: Mainline Flush]
   └── Remove 2" end caps at (500, 245) & (285, 340) -> Run pump at full flow for 5 minutes.
         |
         ▼
[STAGE B: Subline Flush]
   └── Close mainline caps; open all 16 Flush Valves (50mm & 40mm) -> Flush until water runs clear.
         |
         ▼
[STAGE C: Lateral Flush]
   └── Close flush valves; remove 16mm end caps from laterals -> Flush for 3 minutes per zone.
         |
         ▼
[STAGE D: Pressure & Leak Test]
   └── Cap all ends; pressurize system to 2.5 kg/cm² for 60 minutes -> Inspect all 120+ joints.
```

Pressure Test Acceptance Criteria:

1. Zero weeping or dripping at all solvent-welded joints, ball valve unions, and filter flanges.
2. Pressure loss after pump shutoff must not exceed **0.2 kg/cm² in 15 minutes**.
3. All 3,000 inline drippers in Plot A and 768 online drippers in Plot B must show uniform wetting patterns with no dry skips or blowing emitters.

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9. OPERATION & PREVENTATIVE MAINTENANCE

Recommended Daily Irrigation Schedule:

- **Summer (Peak Vegetative & Pod Stage):** 35–45 Liters per plant daily.
    - Plot A (PKM-1): Run 2 Riser Stations simultaneously for 45 minutes; repeat for remaining 2 stations.
    - Plot B (Bhagya): Run 2 Riser Stations simultaneously for 40 minutes (drippers tuned to 35 LPH).
- **Winter / Rainy Season:** 15–20 Liters per plant every 2–3 days.

Routine Maintenance Schedule:

- **Daily:** Open filter drain ball valve for 15 seconds while pump runs to purge trapped silt.
- **Weekly:** Unscrew filter housing, remove disc cartridge, and wash under high-pressure hose.
- **Monthly:** Open all 16 subline flush valves and 80 lateral endcaps for 2 minutes to flush accumulated silt.
- **Seasonal (Every 6 Months):** Acid treatment using 0.5% commercial Hydrochloric Acid (HCl) to dissolve calcium carbonate scaling and fertilizer salts in emitter labyrinths.